06/06/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

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PROJECT PHOENIX FOR RE-NEET-2026

MM: 720 **CST - 06** **Time: 180 Mins.**

Answers

1. (3)	37. (2)	73. (4)	109. (2)	145. (2)
2. (2)	38. (3)	74. (2)	110. (2)	146. (3)
3. (2)	39. (3)	75. (4)	111. (2)	147. (4)
4. (3)	40. (3)	76. (2)	112. (3)	148. (4)
5. (3)	41. (4)	77. (1)	113. (1)	149. (4)
6. (4)	42. (3)	78. (2)	114. (3)	150. (1)
7. (4)	43. (4)	79. (2)	115. (3)	151. (4)
8. (3)	44. (2)	80. (3)	116. (2)	152. (2)
9. (4)	45. (3)	81. (2)	117. (3)	153. (1)
10. (2)	46. (1)	82. (3)	118. (1)	154. (1)
11. (3)	47. (1)	83. (4)	119. (1)	155. (3)
12. (3)	48. (1)	84. (4)	120. (4)	156. (2)
13. (4)	49. (3)	85. (3)	121. (3)	157. (1)
14. (3)	50. (3)	86. (4)	122. (4)	158. (4)
15. (3)	51. (2)	87. (3)	123. (4)	159. (4)
16. (4)	52. (1)	88. (3)	124. (4)	160. (3)
17. (3)	53. (4)	89. (1)	125. (3)	161. (4)
18. (4)	54. (4)	90. (1)	126. (2)	162. (1)
19. (4)	55. (2)	91. (4)	127. (2)	163. (4)
20. (1)	56. (2)	92. (1)	128. (4)	164. (3)
21. (2)	57. (3)	93. (2)	129. (2)	165. (3)
22. (2)	58. (1)	94. (4)	130. (3)	166. (1)
23. (3)	59. (4)	95. (3)	131. (3)	167. (4)
24. (3)	60. (1)	96. (4)	132. (1)	168. (1)
25. (2)	61. (4)	97. (3)	133. (3)	169. (2)
26. (3)	62. (2)	98. (3)	134. (2)	170. (3)
27. (4)	63. (1)	99. (4)	135. (4)	171. (3)
28. (2)	64. (2)	100. (3)	136. (3)	172. (3)
29. (4)	65. (4)	101. (1)	137. (3)	173. (3)
30. (4)	66. (2)	102. (4)	138. (2)	174. (4)
31. (3)	67. (3)	103. (4)	139. (2)	175. (3)
32. (1)	68. (1)	104. (2)	140. (1)	176. (3)
33. (2)	69. (2)	105. (4)	141. (3)	177. (1)
34. (2)	70. (2)	106. (2)	142. (1)	178. (4)
35. (3)	71. (2)	107. (1)	143. (3)	179. (4)
36. (1)	72. (2)	108. (4)	144. (4)	180. (3)

Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:

$R = \frac{V}{I} = \frac{20}{2.5} = 8\ \Omega$

Now $R = \frac{V}{I}$

$\Rightarrow \left(\frac{\Delta R}{R}\right)_{\max} = \left(\frac{\Delta V}{V}\right) + \left(\frac{\Delta I}{I}\right)$

$= \frac{1}{200} + \frac{0.05}{2.5}$

$\Delta R = \frac{2.5}{100} \times 8 = 0.2\ \Omega$

(2) **Answer :** (2)
Solution:

$dw = \vec{F}.d\vec{r} = 8\left(y\hat{i} + x\hat{j}\right).\left(dx\hat{i} + dy\hat{j}\right)$

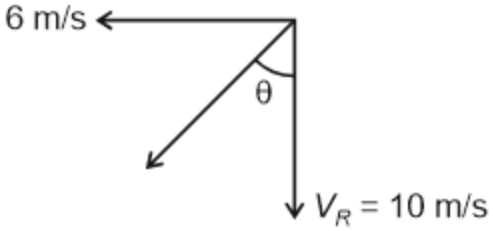
$= 8[xdy + ydx] = 8d(xy)$

$w = 8\int d(xy) = 8\ [xy]_{2,2}^{\sqrt{2},0}$

$= 8[0 - 4] = -32$ J

(3) **Answer :** (2)
Solution:
Area under velocity-time graph gives displacement and in one dimensional motion, constant speed in an interval have zero acceleration.

(4) **Answer :** (3)
Solution:



With respect to man

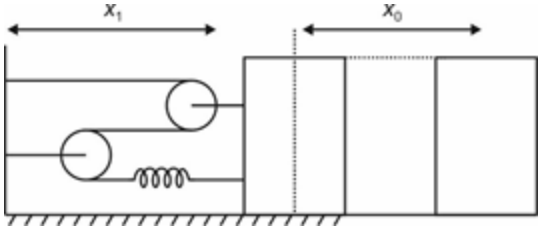
$\tan\theta = \frac{6}{10} = \frac{3}{5}$

$\theta = \tan^{-1}\left(\frac{3}{5}\right)$

(5) **Answer :** (3)
Solution:

$t = \frac{d}{v_{\perp r}} = \frac{400}{10} = 40\ \sec$ [$v_{\perp r}$ → Velocity perpendicular to river velocity]

(6) **Answer :** (4)
Solution:
Total length *l* of string is constant. When block *A* is shifted x_0 to right. Total length of spring and string is



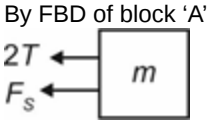
$\lambda = 3(x_1 + x_0)$

Initial length of string and spring = $3x_1$

Extension in spring = $3x_0$

Tension in string = force due to spring.

By FBD of block 'A'



$F_S + 2T = m_A a_A$

$a_A = \frac{9kx_0}{m_A}$

(7) **Answer :** (4)

Solution:

For a moving body distance covered cannot be zero, while displacement of the body will be zero if the body returns to its original position.

(8) **Answer :** (3)

Solution:

Using momentum conservation along horizontal direction, velocity of trolley remain same

(9) **Answer :** (4)

Hint:

$T = mr\omega^2$

Solution:

Tension at distance *x* from pivoted and

$T = \frac{1}{2}\frac{m\omega^2}{l}\left(l^2 - x^2\right)$

$T = \frac{1}{2}\frac{m\omega^2}{l}\left[l^2 - \left(\frac{\sqrt{3}l}{2}\right)^2\right]$

$T = \frac{1}{2}\frac{m\omega^2}{l}\left[\frac{4l^2 - 3l^2}{4}\right]$

$T = \left(\frac{m\omega^2 l}{8}\right)$

(10) **Answer :** (2)

Solution:

$\tau = Mg\frac{L}{2}\cos 60^\circ$

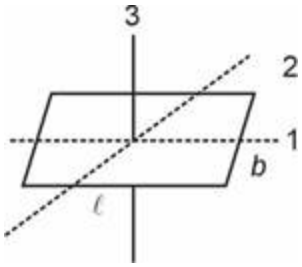
$\frac{ML^2}{3}\alpha = Mg\frac{L}{2}\cos 60^\circ$

$\alpha = \frac{3g\cos 60^\circ}{2L} = \frac{3 \times 10 \times 1}{2 \times 3 \times 2}$

= 2.5 rad/s^2

(11) **Answer :** (3)

Solution:



$I_1 = \frac{Mb^2}{12}$

$I_2 = \frac{Ml^2}{12}$

$I_3 = I_1 + I_2 = \frac{M(l^2 + b^2)}{12}$, using perpendicular axis theorem.

(12) **Answer :** (3)

Solution:

For (a) $I \approx 0$

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For (b) $I = \frac{ML^2}{12}$

For (c) $I = \frac{ML^2}{3}$

∴ For (c) τ is more as *I* is more.

(13) Answer : (4)
Solution:
By standard result.

(14) Answer : (3)
Solution:

$W = \Delta U = k\, q_1\, q_2 \left[\frac{100}{4} - \frac{100}{10}\right]$

$= 9 \times 10^9 \times 54 \times 10^{-12} \times 15$
$= 7.29$ J

(15) Answer : (3)
Solution:

$E_1 = -\frac{GmM}{2\left(R + \frac{R}{3}\right)} = -\frac{3GmM}{8R}$

$E_2 = -\frac{GmM}{2\left(R + R\right)} = -\frac{GmM}{4R}$

Energy required

$= E_2 - E_1 = -\frac{GmM}{4R} + \frac{3}{8}\frac{GmM}{R}$

$= \frac{GmM}{R}\left[-\frac{1}{4} + \frac{3}{8}\right]$

$= \frac{GmM}{R}\left[\frac{-2+3}{8}\right]$

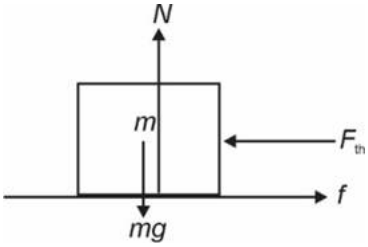
$= \frac{GmM}{8R} = \frac{mgR^2}{8R}$

$= \frac{mgR}{8}$

(16) Answer : (4)
Solution:

Velocity of efflux, $v = \sqrt{2gh}$

Mass flow rate = av



$N = mg$, $F_{th} = f$, $F_{th} = v_r \frac{dm}{dt}$

For minimum μ, $f = f_{s\max}$

$\therefore\ v_r \frac{dm}{dt} = \mu Mg$

$v \times \rho a v = \mu \times \rho A h g$

$\mu = \frac{a \times 2gh}{Ahg}$

$\mu = \frac{2a}{A}$

(17) Answer : (3)
Solution:

Let θ be temperature of interface, then in series heat current must be same.

i.e. $\frac{\Delta\theta_1}{R_1} = \frac{\Delta\theta_2}{R_2}$, $R = \frac{l}{KA}$

$\Rightarrow \frac{2KA}{l}(100 - \theta) = \frac{KA}{l}(\theta - 40)$

$\Rightarrow 200 - 2\theta = \theta - 40$

$240 = 3\theta$

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$\Rightarrow \boxed{\theta = 80^{\circ}\mathrm{C}}$

(18) Answer : (4)
Solution:
$\frac{dQ}{dt} = eA\sigma T^4$

$\frac{E}{E'} = \frac{A}{\frac{A}{2}} \times \frac{\left(127+273\right)^4}{(800)^4} = 2 \times \left(\frac{400}{800}\right)^4$

$\frac{E}{E'} = \frac{1}{2^3} \Rightarrow E' = 8E$

(19) Answer : (4)
Solution:
$I_Z = I_x + I_y$ is applicable for planar bodies and not for three-dimensional bodies.
I = ΣMr^2 In a rigid body, angular velocity of any point w.r.t. any other point is constant.

(20) Answer : (1)
Solution:
$Q = 3\Delta U$
$\Delta U = nC_V\Delta T$
= $\frac{nR\Delta T}{\frac{5}{3}-1}$
= $\frac{3}{2}nR\Delta T$
nCΔT = $3 \times \frac{3}{2}nR\Delta T$
$C = \frac{9R}{2}$

(21) Answer : (2)
Solution:
$P = \frac{\alpha T^2}{V}$
$PV = \alpha T^2$
$\int PdV = \int_{T_0}^{3T_0} 2\alpha T dT$
$W = \alpha\left[T^2\right]_{T_0}^{3T_0}$
$W = 8\alpha T_0^2$

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(22) Answer : (2)
Solution:
$T = \frac{2\pi}{\omega}$
$= (V_p)_{\max} = A\omega = 20 \times 10^{-2} \times 200$
= 40 m/s

(23) Answer : (3)
Solution:
$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$
$\sqrt{3} = \frac{\sin\left(\frac{60^\circ+\delta_m}{2}\right)}{\sin\left(\frac{60^\circ}{2}\right)}$
$\frac{\sqrt{3}}{2} = \sin\left(30^\circ + \frac{\delta_m}{2}\right) = \sin 60^\circ$
$60^\circ = 30^\circ + \frac{\delta_m}{2}$
δ_m = 60°

(24) Answer : (3)
Solution: $\mathrm{KE} = \frac{3}{4}\ \mathrm{T.E} \Rightarrow \frac{1}{2}m\omega^2\left(A^2 - x^2\right) = \frac{3}{4} \times \frac{1}{2}m\omega^2A^2$

So, $x = \frac{A}{2}$

(25) Answer : (2)
Solution:

For translational equilibrium ⇒ $\Sigma\vec{F} = 0$

For rotational equilibrium ⇒ $\Sigma\vec{\tau} = 0$

$\tau = I\alpha$

$I_{\text{solid sphere}} = \frac{7}{5}MR^2$

(26) Answer : (3)
Solution:

We know that $X_C = \frac{1}{\omega C}$ and $X_L = \omega L$

Also, at $\omega = \omega_r$: $X_L = X_C$

⇒ For $\omega < \omega_r$: capacitive

and At $\omega = \omega_r : z = \sqrt{R^2 + (X_L - X_C)^2} = R$

(27) Answer : (4)
Hint:

$$\frac{\text{Voltage across primary coil}}{\text{Voltage across secondary coil}} = \frac{\text{No. of turns in primary coil}}{\text{No. of turns in secondary coil}}$$

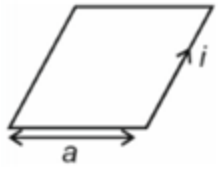
Solution:

$V_2 = \left(\frac{N_2}{N_1}\right)V_1$

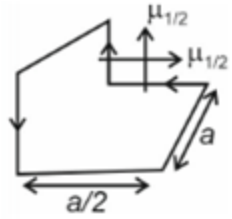
$V_2 = \left(\frac{3000}{1000}\right)100$

= 300 V

(28) Answer : (2)
Solution:



$\left(\vec{\mu}_1\right) = ia^2$



$\left(\vec{\mu}_2\right) = \sqrt{\left(\frac{\mu_1}{2}\right)^2 + \left(\frac{\mu_1}{2}\right)^2} = \frac{\mu_1}{\sqrt{2}}$

$\left(\frac{\vec{\mu}_1}{\mu_2}\right) = \sqrt{2}$

(29) Answer : (4)
Solution:

Displacement current exists within the gap of a capacitor when the electric field between its plates is changing, which typically happens when the charge on the capacitor is changing.

(30) Answer : (4)
Solution:

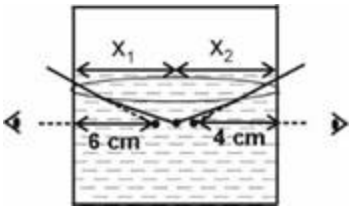
The potential difference appears across the two ends if $\vec{V}$, $\vec{B}$ and l are mutually perpendicular to each other.

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(31) **Answer :** (3)
Solution:
$V' = n^{2/3}V = 8^{2/3} \times 2$ volt
= 4 × 2 volt
= 8 volt

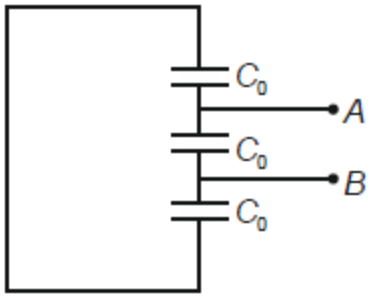
(32) **Answer :** (1)
Solution:
$B = \frac{\mu_0 i}{4\pi R} \times \theta = \frac{\mu_0 i}{4\pi R} \times \frac{5\pi}{3}$

(33) **Answer :** (2)
Solution:



$6 = \frac{x_1}{\mu}$
$\Rightarrow x_1 = 6\mu$ and $x_2 = 4\mu$
$t = x_1 + x_2 = 10 \times \frac{3}{2} = 15$ cm

(34) **Answer :** (2)
Solution:



As $C = C_0 + \frac{C_0}{2} = \frac{3C_0}{2} = \frac{3}{2}\frac{\varepsilon_0 A}{d}$

(35) **Answer :** (3)
Hint:
Use Boolean algebra.
Solution:
$Y = \overline{\overline{A.B} + \overline{A.B}}$
$Y = \overline{\overline{A.B}}.\ \overline{\overline{A.B}}$
$Y = (A.B).\ (A.B)$
$\boxed{Y = A.B}$

(36) **Answer :** (1)
Solution:
Optical path = μx
= 2 × 8 = 16 m

(37) **Answer :** (2)
Solution:
$\beta = \frac{\lambda D}{d}$
Let length of segment of screen = l
$\Rightarrow\ l = 8\beta_1 = \frac{8\lambda_1 D}{d}$ ...(1)
and $l = n\beta_2 = \frac{n\lambda_2 D}{d}$ ...(2)
from (1) and (2)
$8\lambda_1 = n\lambda_2$
8(600 nm) = n(400 nm)

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n = 12

(38) Answer : (3)
Solution:
$y_n = \frac{nD\lambda}{d}$,
$y_n = \frac{nD\lambda}{d}$, $5 \times 10^{-3} = \frac{3D\lambda}{d} - \frac{D\lambda}{d}$
$\Rightarrow d = 2 \times 10^{-4}$ m

(39) Answer : (3)
Hint:
Use de-Broglie wavelength $\lambda = \frac{h}{p}$
and $p = \sqrt{2mK} = \sqrt{2mqV}$.
Solution:
de-Broglie wavelength
$\lambda = \frac{h}{\sqrt{2mqV}}$
$\lambda \propto \frac{1}{\sqrt{V}}$
$\therefore \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{V_1}{V_2}}$
$\frac{\lambda_2}{1} = \sqrt{\frac{130}{520}}$
$\lambda_2 = \frac{1}{2}$
= 0.5 Å

(40) Answer : (3)
Solution:
$w = h\nu_{\text{th}} = \frac{6.6\times10^{-34}\times2\times10^{14}}{1.6\times10^{-19}}$ eV
w = 0.8 eV

(41) Answer : (4)
Solution:
$\mu = a + \frac{b}{\lambda^2} + ...$ and $\theta_C = \sin^{-1}\left(\frac{1}{\mu}\right)$
So, orange light and red light undergo refraction and emerge in air.
∴ Violet, indigo, blue and green light show total internal reflection.

(42) Answer : (3)
Solution:
x = M.S.R + n × L.C
= 1 cm + 25 × 0.001 cm
= 1 cm + 0.025
=1.025 cm

(43) Answer : (4)
Solution:
$I_R = I_1 + I_2 - 2\sqrt{I_1 I_2}$
$= 9I_0 + 4I_0 - 2\sqrt{36I_0^2}$
$= 13I_0 - 12I_0$
$= I_0$

(44) Answer : (2)
Solution:
$\frac{1}{2}mv^2 = \frac{12400}{4000} - 1.2$
$\frac{1}{2}mv^2 = 3.1 - 1.2 = 1.9$ eV
$v^2 = \frac{2\times1.9\times1.6\times10^{-19}}{9.1\times10^{-31}}$
v = 8.17 × 10^5 m/s

(45) Answer : (3)
Solution:
Binding energy of $^2_1$H= 1.15 × number of nucleons

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= 1.15 × 2 = 2.3 MeV
Total binding energy (Reactants)
= 2.3 + 2.3 = 4.6 MeV
Total binding energy of $^{4}_{2}He$= 7.1 × 4 = 28.4 MeV
⇒ Q = 28.4 – 4.6 = 23.8 MeV

CHEMISTRY

(46) Answer : (1)
Solution:
a. 18 g $C_6H_{12}O_6$
Number of oxygen atoms
= $\frac{18}{180} \times 6 \times N_A$
= 0.6 N_A
b. 44.8 L NO_2 gas at STP
Number of oxygen atoms
= $\frac{44.8}{22.4} \times 2 \times N_A = 4\,N_A$
c. 1.225 g $KClO_3$
Number of oxygen atoms
$= \frac{1.225}{122.5} \times 3 \times N_A = 0.03\,N_A$
d. 11.2 L CO gas at STP
Number of oxygen atoms
$= \frac{11.2}{22.4} \times 1 \times N_A = 0.5\,N_A$

(47) Answer : (1)
Solution:
Weight of mixture of BaO and MCO_3 = 4.08 g

$MCO_3 \xrightarrow{\Delta} MO + CO_2$
Mass of BaO + MO = 3.64 g
Mass of CO_2 evolved = 4.08 – 3.64 = 0.44 g
moles of CO_2 = $\frac{0.44}{44} = 10^{-2} = 10\text{ m mol}$
mEq of CO_2 = 10 × 2 = 20 mEq of CO_2
For acid = 100 × 1 = 100 mEq
Excess acid = 100 – 16 × 2.5 = 60 mEq
Eq. of w of (BaO) = $\frac{Mw}{2}$
mEq of BaO = 40 mEq = $\frac{40\times10^{-3}\times154}{2} = 3.08\,g$
Weight of MO = 3.64 – 3.08 = 0.56 g
20 mEq of MO = 0.56
Equivalent weight of MO = $\frac{0.56\times1000}{20} = 28$
Molecular weight = 28 × 2 = 56
Atomic weight of M = 56 – 16 = 40
Metal is Ca.

(48) Answer : (1)
Solution:

Energies of the orbitals in the same subshell decrease with increase in the atomic number $\left(Z_{eff}\right)$.

So correct order of energies of 2s orbitals is
$E_{2S}(H) > E_{2S}(Li) > E_{2S}(Na) > E_{2S}(K)$

(49) Answer : (3)
Solution:

$E_n = -13.6 \times \frac{Z^2}{n^2}$

$-7.65 = -13.6 \times \frac{9}{n^2}$

$n = 4$

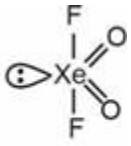
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(50) **Answer :** (3)
Solution:
Since I.E. of Na is 496 kJ/mol.
So (n + 2) should be 11
$n+2 = 11$
So, n = 9

(51) **Answer :** (2)
Solution:



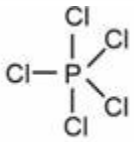
Square planar



See-saw



Square pyramidal



Trigonal bipyramidal

(52) **Answer :** (1)
Solution:
Bond length $\propto \frac{1}{\text{Bond order}}$
Bond order of $O_2^+ = 2.5$
Bond order of $O_2 = 2$
Bond order of $O_2^- = 1.5$
$\therefore$ The order of bond length is $O_2^+ < O_2 < O_2^-$

(53) **Answer :** (4)
Solution:
C → A; Reversible isothermal compression.
W = $-2.303nRT\log\frac{V_2}{V_1}$
= $-2.303 \times 3 \times 2 \times 300 \times \log\frac{30}{60}$
= $2.303 \times 1800 \times 0.3$
= 1.24 kcal
ΔH = 0, for cyclic process

(54) **Answer :** (4)
Solution:
$60 = \frac{\text{weight of phosphorus}}{0.31 \times 1000} \times 100$
Weight of phosphorus = 3.1 × 60 = 186 g
40% of phosphorus is converted into PCl_3
$186 \times \frac{40}{100} = 74.4$ g
Energy released = $\frac{x}{31} \times 74.4 = 2.4x$
60% of phosphorus converted into PCl_5
186 – 74.4 = 111.6 g
Energy released = $\frac{y}{31} \times 111.6 = 3.6y$
Total energy released = 2.4 x + 3.6 y

(55) **Answer :** (2)
Solution:
$Ag_2CrO_4(s) \rightleftharpoons \underset{2s}{2Ag^+} + \underset{s+0.2 \text{ - } 0.2}{CrO_4^{2-}}$

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$\underset{-}{K_2CrO_4} \rightarrow \underset{0.4M}{2K^+} + \underset{0.2\ M}{CrO_4^{2-}}$

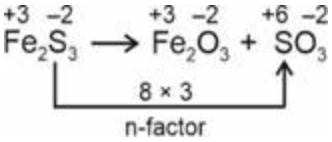
$K_{sp} = 1.1 \times 10^{-12} = (2s)^2 \times (0.2)$

$s^2 = \frac{1.1\times10^{-12}}{0.2\times4}$

$s = 1.17 \times 10^{-6}$ mol/L

(56) Answer : (2)
Solution:
• CH_3COONa undergoes anionic hydrolysis hence solution will be basic in nature; pH > 7.
• CH_3NH_3Cl undergoes Cationic hydrolysis hence solution will be acidic in nature; pH < 7.
• Water is neutral and at high temperature pH < 7.
• NaCl does not undergo hydrolysis so neutral with pH = 7.

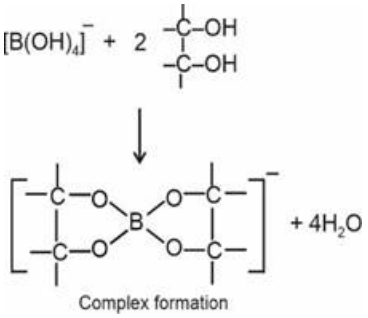
(57) Answer : (3)
Solution:



Number of electron lost per molecule (n) = 24
Equivalent mass = $\frac{M}{n} = \frac{M}{24}$

(58) Answer : (1)
Solution:
$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

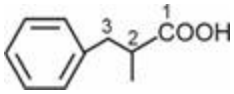
(59) Answer : (4)
Solution:
H_3BO_3 does not undergo self-ionization.



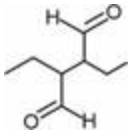
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(60) Answer : (1)
Solution:
Producer gas $\rightarrow$ CO + N_2
Synthesis gas $\rightarrow$ CO + H_2
Dry ice $\rightarrow$ Solid CO_2
Silica $\rightarrow$ SiO_2

(61) Answer : (4)
Solution:



2-Methyl-3-phenylpropanoic acid



2,3-Diethylbutan-1,4-dial

$\overset{6}{HC} \equiv \overset{5}{C} - \overset{4}{CH_2} - \overset{3}{CH_2} - \overset{2}{CH} = \overset{1}{CH_2}$ Hex-1-en-5-yne

(62) Answer : (2)
Solution:
Presence of H-atom at α carbon to carbonyl group can lead to keto-enol tautomerism. H atom of bridgehead carbon cannot be involved in keto enol tautomerism.

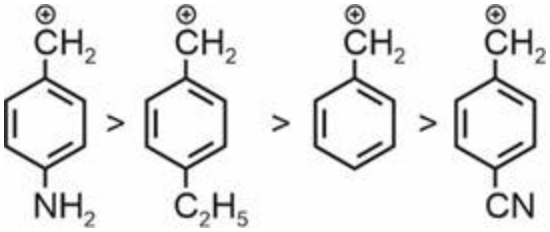
(63) Answer : (1)
Solution:
Experimental values at STP,
V_1 = 50 mL, V_2 = ?
T_1 = 300 K, T_2 = 273 K
P_1 = 720 – 20 = 700 mmHg, P_2 = 760 mmHg

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow V_2 = \frac{P_1V_1T_2}{T_1P_2} = \frac{700\times50\times273}{300\times760} = 41.9\,\text{mL}$$

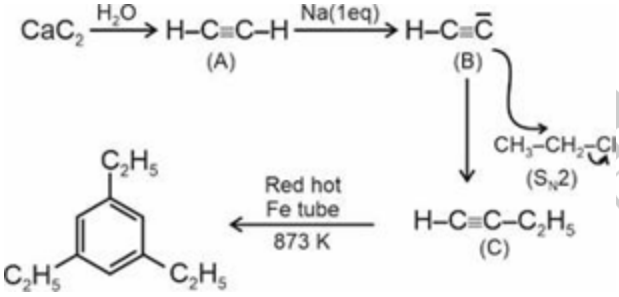
$$\%\ \text{of nitrogen} = \frac{1}{8} \times \frac{\text{Volume of } N_2 \text{ at S.T.P}}{\text{Weight of unknown compound}}$$

$$= \frac{1}{8} \times \frac{41.9}{0.5} = 10.47\%$$

(64) Answer : (2)
Hint:
$–NH_2$ group present at para-position in benzyl carbocation stabilises it by +R effect.
Solution:



(65) Answer : (4)
Hint:
Ethyne is prepared by treating calcium carbide with water.
Solution:



(66) Answer : (2)
Solution:
NaCl → i = 1 + α = 1 + 0.9 = 1.9
$CaCl_2$ → i = 1 + 2α = 1 + 2 × 0.6 = 2.2
$AlCl_3$ → i = 1 + 3α = 1 + 3 × 0.5 = 2.5
$K_3[Fe(CN)_6]$ → i = 1 + 3α = 1 + 3 × 0.75 = 3.25

(67) Answer : (3)
Solution:
Calculated total pressure P_T
= $X_xP_x^o + X_yP_y^o$
= $\frac{5}{15}\times63 + \frac{10}{15}\times78$
= 21 + 52 = 73 torr
As observed P_T is greater than calculated, P_T
• It shows positive deviation.
• X–Y interaction is smaller than X–X and Y–Y interaction.
• Mixing is endothermic.

(68) Answer : (1)

Hint:

Anode: $H_2O(l) \rightarrow \frac{1}{2}O_2(g)+2H^+(aq)+2e^-$

Cathode : $Ag^+(aq) + e^- \rightarrow Ag(s)$

Solution:

$w = \frac{Eit}{96500} \Rightarrow$ Mole x n-factor = $\frac{it}{96500}$

Mole (O_2) × 4 = $\frac{9.65\times60\times60}{96500} = 0.36$

Mol(O_2) = 0.09

Volume of O_2 (g) at STP = 0.09 × 22.4 = 2.016 L

(69) Answer : (2)

Solution:

pH = 14

$\therefore$ pOH = 0

$\therefore [OH^-] = 1$ M

$[Cu^{2+}][OH^-]^2 = 1 \times 10^{-19}$

$\therefore [Cu^{2+}] = 1 \times 10^{-19}$

$E_{Cu^{2+}/Cu} = 0.34 - \frac{0.059}{2}\log 10^{19} = -0.22\,V$

(70) Answer : (2)

Solution:

$K = \frac{0.693}{30}$ min^{-1}

$Kt = 2.303\log\frac{a}{a-x}$

$\frac{0.693\times70}{30\times2.303} = \log\frac{a}{a-x}$

$\frac{a}{a-x} = \text{antilog}\,0.7021$

$\frac{a}{a-x} = 5.036$

$\Rightarrow \frac{a-x}{a} = \frac{1}{5.036} \simeq 0.2$

(71) Answer : (2)

Solution:

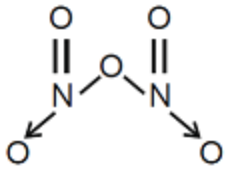
Compounds	Melting point (K)
H_2Te	222
H_2S	188
H_2O	273

(72) Answer : (2)

Solution:

$4\,HNO_3 \xrightarrow{P_4O_{10}} 4\,HPO_3 + \underset{(A)}{2N_2O_5}$

N_2O_5 :



(73) Answer : (4)

Solution:

Cr, Mo and W have highest melting point in 3*d*, 4*d* and 5*d* series respectively.

(74) Answer : (2)

Solution:

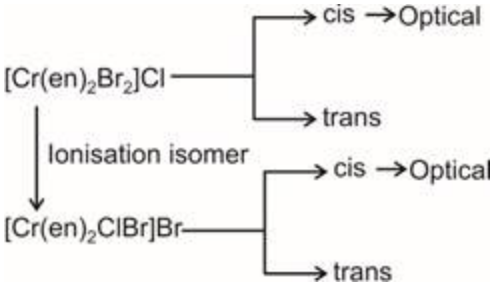
$Cr_2O_7^{2-} \underset{pH<7}{\overset{pH>7}{\rightleftharpoons}} CrO_4^{2-}$

(75) Answer : (4)

Solution:

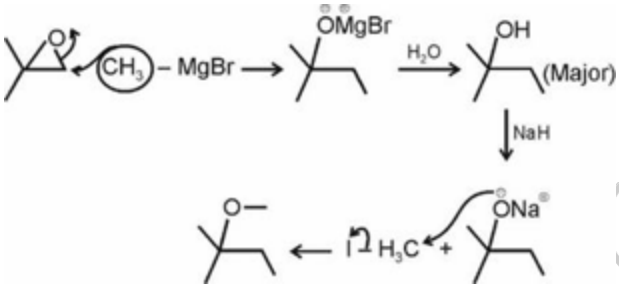
If a substance absorbs light in the yellow region, it reflects or transmits its complementary colour, which is violet.

(76) Answer : (2)
Solution:

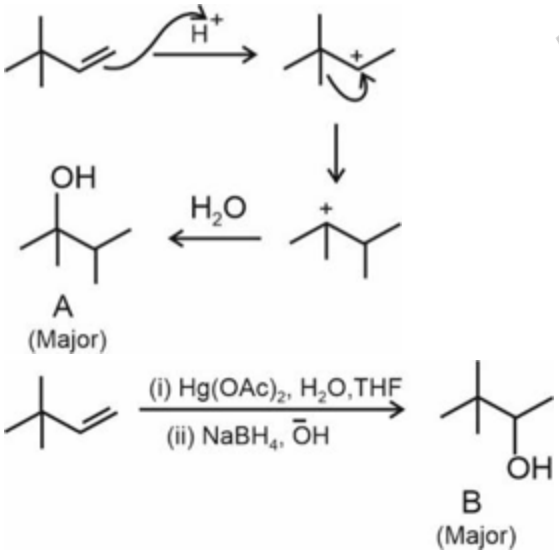


(77) Answer : (1)
Solution:
Due to the presence of positive charge, synergic bonding from metal to carbonyl ligand will decrease. Hence, M – C bond length will increase and C – O bond length will decrease.

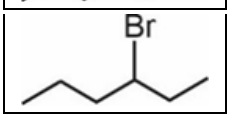
(78) Answer : (2)
Hint:
Grignard reagent preferably attacks by S_N2 mechanism at sterically less hindered carbon of the epoxide.
Solution:



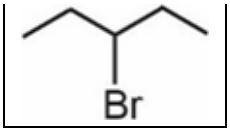
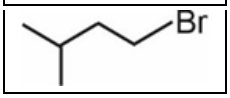
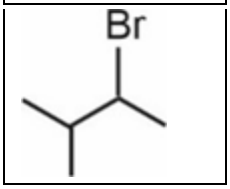
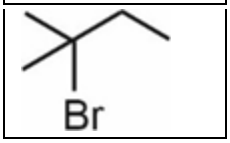
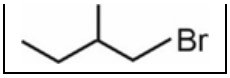
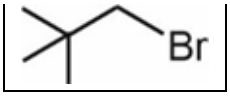
(79) Answer : (2)
Solution:



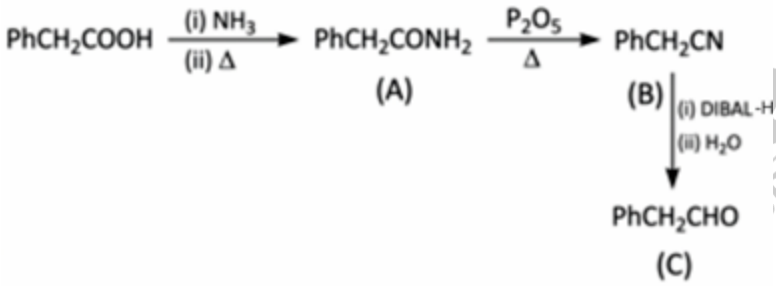
(80) Answer : (3)
Solution:

	1-Bromopentane
	2-Bromopentane
	3-Bromopentane

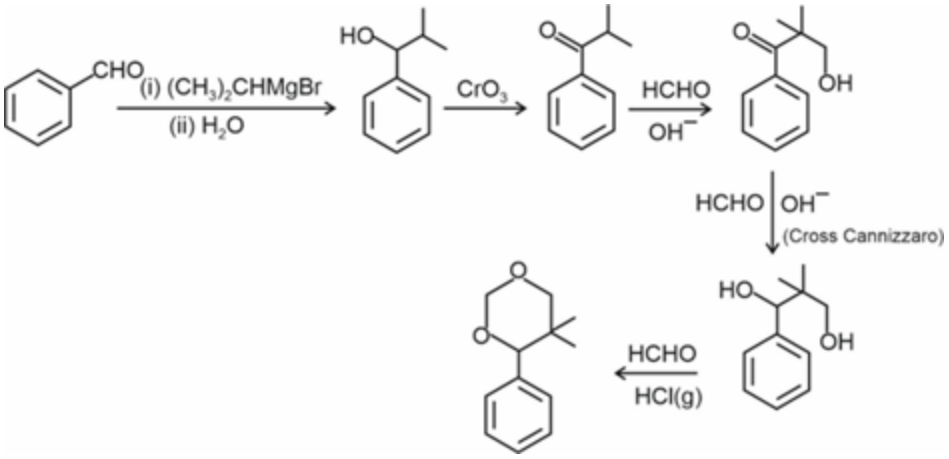
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	1-Bromo-3-methylbutane
	2-Bromo-3-methylbutane
	2-Bromo-2-methylbutane
	1-Bromo-2-methylbutane
	1-Bromo,2-2-dimethylpropane

(81) Answer : (2)
Solution:



(82) Answer : (3)
Solution:

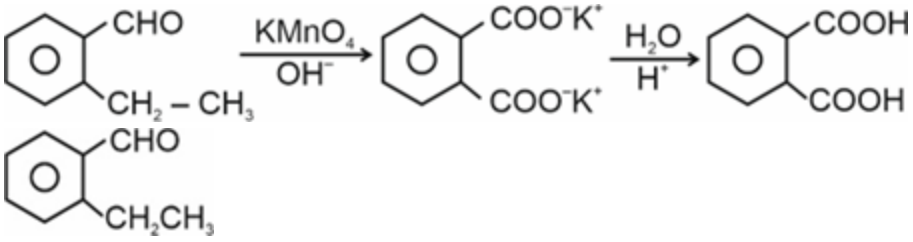


(83) Answer : (4)
Hint:
For aldol condensation reaction, 2α hydrogen atoms are required.
Solution:
2-Methylpropanal contains only one α-hydrogen atom hence it does not undergo aldol condensation reaction.

(84) **Answer :** (4)
Solution:

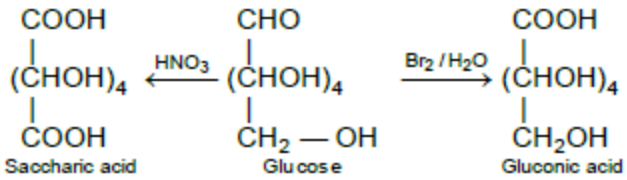
Mn^{2+} belongs to group-IV while Sr^{2+} belongs to group-V.

(85) **Answer :** (3)
Solution:

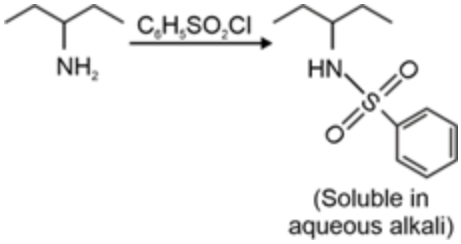


gives 2,4-DNP derivative and reduces Tollens' reagent. The compound lacks α-H, so, it can undergo Cannizzaro reaction also.

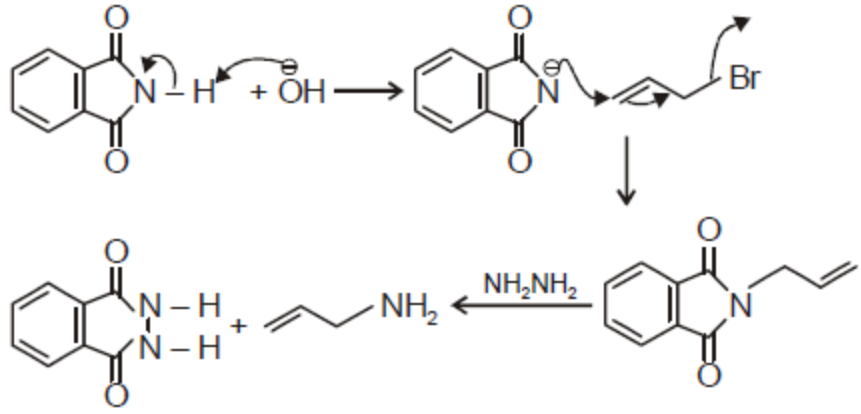
(86) **Answer :** (4)
Solution:



(87) **Answer :** (3)
Solution:



(88) **Answer :** (3)
Solution:



(89) **Answer :** (1)
Solution:

$NH_2\,COONH_4(s) \rightleftharpoons 2\,NH_3(g) + CO_2(g)$

2P + P = 3P (initial total pressure)

Initial, $K_p = (2P)^2 (P) = 4P^3$

After addition of NH_3 ⇒ $P_{NH_3} = \frac{3}{2}(3P) = \frac{9}{2}P$

Let $P_{CO_2} = x$ then $\left(\frac{9}{2}P\right)^2\left(x\right) = 4P^3$

$\left(\frac{81P^2}{4}\right)\left(x\right) = 4P^3$

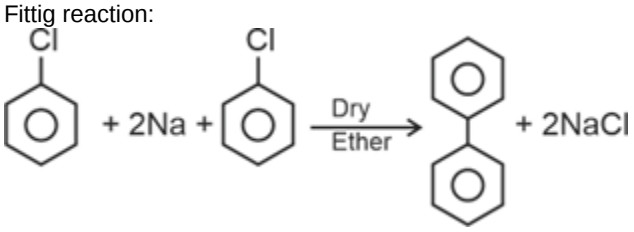
$x = \frac{16P^3}{81P^2} = \frac{16}{81}P$

$\frac{\text{Final total P}}{\text{Initial total P}} = \frac{\frac{9}{2}P + \frac{16}{81}P}{\left(\frac{3P}{1}\right)} = \frac{761}{486}$

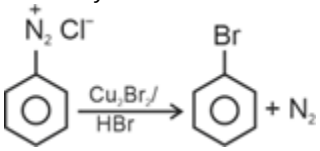
(90) **Answer :** (1)
Solution:
Swarts reaction
$CH_3Br + AgF \rightarrow CH_3F + AgBr$
Finkelstein reaction

$CH_3Cl + NaI \xrightarrow[\text{acetone}]{\text{dry}} CH_3I + NaCl$

Fittig reaction:



Sandmeyer's reaction:



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BIOLOGY

(91) **Answer :** (4)
Hint:
Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.
Solution:
When we go higher from species to kingdom, problem of classification becomes more complex.

(92) **Answer :** (1)
Solution:
Invertase enzyme catalyzes the conversion of sucrose into glucose and fructose

(93) **Answer :** (2)
Hint:
Mucor belongs to the class Phycomycetes, whereas, *Ustilago* belongs to the class Basidiomycetes.
Solution:
Dikaryophase is absent in *Mucor*. Biological classification of plants and animals was first proposed by Aristotle.

(94) **Answer :** (4)
Hint:
Linnaeus classification of angiosperm is a sexual system of classification.
Solution:
He classified angiosperms on the basis of androecium structure.

(95) **Answer :** (3)
Solution:
• Complex-IV refers to cytochrome-c oxidase complex containing cytochromes a and a_3 and two copper centres.
• Cytochrome bc_1 complex is complex-III.

(96) **Answer :** (4)

Solution:
Sorghum, maize and sugarcane are C_4 plants. Tomato is a C_3 plant.

(97) Answer : (3)
Hint:
The given organism is a dividing bacterium
Solution:
'X' represents the non-cellulosic cell wall.
'Y' represents the cell membrane.
'Z' represents the circular DNA.

(98) Answer : (3)
Solution:
Flowers of *Vallisneria* are water pollinated.

(99) Answer : (4)
Hint:
Mesophyll is not differentiated into palisade and spongy parenchyma in monocots.
Solution:
There is no palisade parenchyma in the leaves of *Triticum aestivum* (wheat).

(100) Answer : (3)
Hint:
Absolute growth rate is the measurement and comparison of the total growth per unit time.
Solution:
The relative growth rate is calculated by:

$$\frac{Growth\ per\ unit\ time}{Initial\ size} \times 100$$

The relative growth rate of leaf A in first, second and third years, respectively are 100%, 100% and 30%.
The relative growth rate of leaf B in first, second and third year, respectively are 100%, 25% and 40%.

(101) Answer : (1)
Solution:
G_2 – 40 pg Meiosis I – 20 pg
Meiosis II – 10 pg Egg cell – 10 pg

(102) Answer : (4)
Solution:
In a double stranded DNA, G+C/A+T base ratio is species specific and vary species to species.

(103) Answer : (4)
Solution:
Root hairs are present in the region of maturation

(104) Answer : (2)
Hint:
Only 10% of the energy is transferred to each trophic level from the lower trophic level.
Solution:

$$\text{Grass} \xrightarrow{10\text{ J}} \text{Grasshopper} \xrightarrow{1\text{ J}} \text{Frog} \xrightarrow{0.1\text{ J}} \text{Snake} \xrightarrow{0.01\text{ J}}$$

(105) Answer : (4)
Solution:
According to the five kingdom classification by R. H. Whittaker, both autotrophic and heterotrophic mode of nutrition is found in the kingdom Monera and kingdom Protista.

(106) Answer : (2)
Solution:
Fibroblasts are the matrix and collagen forming cells.
Mast cells – Release vasodilator histamine
Adipocyte – Has peripheral nucleus due to fat globule present in centre of cell.

(107) Answer : (1)
Solution:
Excess nutrients are stored as fats in adipose tissue.

(108) Answer : (4)
Solution:

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In quaternary structure of protein, more than one polypeptide chains are present, *eg*., haemoglobin. Normally, the protein which contains only single polypeptide can possess all the three structures, primary, secondary and tertiary.

(109) Answer : (2)

Hint:

Exclude amino acids.

Solution:

Aspartic acid and alanine are amino acids. Malonate is the competitive inhibitor of succinic dehydrogenase. Succinate is the substrate for succinic dehydrogenase.

(110) Answer : (2)

Solution:

In nucleic acid, the bond between the phosphate and hydroxyl group of sugar is an ester bond.

(111) Answer : (2)

Hint:

The substituents occupy four valency positions on α-carbon.

Solution:

Amino acids are organic compounds. They are substituted methanes. There are four substituent groups occupying the four valency positions of α-carbon. These are hydrogen, carboxyl group, amino group and a variable 'R' group of α-carbon.

(112) Answer : (3)

Hint:

More than one factors influence mechanism of breathing.

Solution:

Mechanism of breathing vary among different groups of animals depending mainly on their habitats and levels of organisation.

(113) Answer : (1)

Hint:

Zn^{2+} is the co-factor of this enzyme.

Solution:

$$CO_2 + H_2O \overset{\text{Carbonic anhydrase}}{\rightleftharpoons} H_2CO_3 \overset{\text{Carbonic anhydrase}}{\rightleftharpoons} HCO_3^- + H^+$$

(114) Answer : (3)

Hint:

Function performed by alveoli

Solution:

The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part whereas the alveoli and their ducts form the respiratory or exchange part of the respiratory system.

The conducting part transports the atmospheric air to the alveoli, clears it from foreign particles, humidifies and also brings the air to body temperature. Exchange part is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

(115) Answer : (3)

Hint:

Animal with closed type of circulation

Solution:

Porifers (*Sycon*) and coelenterates (*Hydra* and *Obelia*), circulate water from their surroundings through their body cavities to facilitate the cells to exchange useful substances. Annelids (*Pheretima*) use special fluid called blood for this purpose.

(116) Answer : (2)

Hint:

Salvinia is a water fern

Solution:

Salvinia lack strobili or cones, they belong to pteridophytes. Most of its members are homosporous, though *Salvinia* is heterosporous.

(117) Answer : (3)

Solution:

As we move from low to high latitude, biodiversity decreases.

(118) Answer : (1)

Solution:

In-situ conservation – National park

Ex-situ conservation – Zoological park, botanical garden, wildlife safari park

(119) Answer : (1)

Hint:

If GPP is equal to the rate of respiration, then GPP = R

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Solution:
We know,
NPP = GPP – R
So, NPP would be zero

(120) Answer : (4)
Hint:
If both strands act as template, they would code for RNA molecules with different sequences.
Solution:
If both RNA molecules code for proteins, the sequence of amino acids in the proteins would be different. Hence, one segment of the DNA would be coding for two different proteins, and this would complicate the genetic information transfer machinery. Second, the two RNA molecules if produced simultaneously would be complimentary to each other, hence would form a double stranded RNA. This would prevent RNA from being translated into protein and the exercise of transcription would become a futile one.

(121) Answer : (3)
Hint:
Red algae shows complex post fertilization development.
Solution:
Members of rhodophyceae (red algae) are deepest dweller, lack flagella throughout their life cycle and their stored food is floridean starch (structurally similar to amylopectin and glycogen).

(122) Answer : (4)
Hint:
Bulliform cells help in the rolling of leaves inwards to minimise water loss.
Solution:
Bulliform cells are present on the adaxial surface of leaves of grasses.

(123) Answer : (4)
Solution:
GAs activate the amylase enzyme and helps in breaking of dormancy of seeds

(124) Answer : (4)
Solution:
The technology of biogas production was developed in India mainly due to the efforts of Indian Agricultural Research Institute (IARI) and Khadi and Village Industries Commission (KVIC).

(125) Answer : (3)
Hint:
Prokaryotic cells show the presence of fimbriae.
Solution:
The middle layer of the cell envelope (cell wall) in the prokaryotic cell prevents these cells from bursting or collapsing. Glycocalyx differs in composition and thickness in different forms.

(126) Answer : (2)
Hint:
The size of a typical bacterium is generally 3-5 μm.
Solution:
Mycoplasma, the smallest cell, is only 0.3 μm long and it lacks a nuclear membrane.

(127) Answer : (2)
Hint:
The network of tiny tubular structures scattered in the cytoplasm represents ER.
Solution:
The biomolecules synthesized in ER reach the *cis* face of Golgi complex. Within the Golgi complex the biomolecules is transported to the *trans* face and ultimately release from this concave face.

(128) Answer : (4)
Solution:
In general, herbivores and plants appear to be more adversely affected by competition than carnivores. If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lack of food. This is the reason why predators in nature are ‘prudent’.

(129) Answer : (2)
Solution:
K in Verhulst-pearl logistic growth represents carrying capacity.

(130) Answer : (3)
Hint:
Datura belongs to Solanaceae family.

Solution:
Solanaceae members can have endospermic seed and berry or capsule fruits.

(131) Answer : (3)
Hint:
Cardiac arrest means complete stoppage of heart beat
Solution:
Heart attack/Myocardial infarction: Occurs when the heart muscles are suddenly damaged by an inadequate blood supply.
Cardiac arrest: It means complete stoppage of heart beat *i.e.*, when the heart stops beating.
Coronary Artery Disease (CAD) : Often referred to as atherosclerosis, affects the vessels that supply blood to the heart muscles.

(132) Answer : (1)
Hint:
70 – 80% of electrolytes and water are reabsorbed by PCT.
Solution:
PCT: All of the essential nutrients, and 70–80% of electrolytes and water are reabsorbed by this segment.
Henle's loop: The descending limb of loop of Henle is permeable to water but almost impermeable to electrolytes. The ascending limb is impermeable to water but allows transport of electrolytes actively or passively.
DCT: Capable of reabsorption of HCO_3^- and selective secretion of hydrogen and potassium ions.

(133) Answer : (3)
Hint:
ADH increases blood pressure
Solution:
ADH can also affect the kidney function by its constrictory effect on blood vessels. This causes an increase in blood pressure. An increase in blood pressure can increase the glomerular blood flow and thereby the GFR.

(134) Answer : (2)
Hint:
Each limb is made of 30 bones.
Solution:
In an adult human, skeletal system is made up of 206 bones. Axial skeleton comprises of 80 bones distributed along the main axis of the body. The bones of the limbs along with their girdles constitute the appendicular skeleton. It comprises of 126 bones. Each limb is made up of 30 bones.

(135) Answer : (4)
Solution:
Heart wall is composed of involuntary striated muscle fibres.

(136) Answer : (3)
Solution:
Zygomatic, clavicle and parietal are paired bones.

(137) Answer : (3)
Hint:
This hormone requires iodine for its synthesis.
Solution:
Deficiency of thyroxine during intrauterine life causes cretinism characterised by stunted growth, mental retardation and deaf mutism.

(138) Answer : (2)
Hint:
Identify a steroid hormone.
Solution:
Norepinephrine, oxytocin and TCT are lipid insoluble hormones hence, they cannot cross the lipid bilayer. They execute their action through extracellular receptors. Estrogen is lipid soluble, steroid hormone which acts through intracellular receptors.

(139) Answer : (2)
Hint:
More than two are correct.
Solution:
Apis, Bombyx and *Pinctada* are economically beneficial invertebrates.
Culex (vector of disease) and *Locusta* (gregarious pest) are harmful insects.

(140) Answer : (1)
Solution:
Those cnidarians which exist in both polyp and medusa forms exhibit alternation of generation (Metagenesis).

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(141) Answer : (3)
Solution:
Crop is essential for the storage of food substances and gizzard helps in crushing of food.

(142) Answer : (1)
Solution:
'X' is 100-150 yellow-coloured filamentous malpighian tubules present at the junction of midgut and hindgut.

(143) Answer : (3)
Solution:
Hint : Arthrodial membrane
Sol. : The body of cockroach is divided into head, thorax and abdomen. In each segment, hardened plates are present as exoskeleton which are joined together by arthrodial membrane. Head is freely movable due to flexible neck.

(144) Answer : (4)
Hint:
Exclude branchial respiration
Solution:
Frogs respire on land and in water by two different methods. In water, skin acts as aquatic respiratory organ (cutaneous respiration). Dissolved oxygen in water is exchanged through the skin by diffusion.
On land, buccal cavity, skin and lungs act as the respiratory organs.

(145) Answer : (2)
Solution:
The cells of alveoli secrete milk which is stored in the cavity of alveoli.
Areola is the pigmented part around nipple.

(146) Answer : (3)
Solution:
Breakdown of the endometrium occurs during the menstrual phase.

(147) Answer : (4)
Solution:
Embryos with more than 8 blastomeres are transferred into the uterus (IUT-intra uterine transfer).

(148) Answer : (4)
Solution:
In vasectomy, a small part of the vas deferens is removed or tied up through a small incision on the scrotum.

(149) Answer : (4)
Solution:
Hint : Theory of biogenesis.
Sol. : Options (1), (2) and (3) are connotations of theory special creation but option (4) is associated with theory of biogenesis which was given by Louis Pasteur.

(150) Answer : (1)
Solution:
Hint : Adaptive radiation.
Sol. : Darwin's finches show adaptive divergence and adaptive radiation.

(151) Answer : (4)
Hint:
Glands that produce saliva
Solution:
Plasmodium (Protozoan) is the causative agent of malaria.
The female *Anopheles* mosquito takes up the gametocytes of *Plasmodium* with the blood meal. Fertilization and development take place in the mosquito's gut. Mature infective stages escape from the gut and migrate to the mosquito's salivary glands. When the mosquito bites another human, sporozoites are injected with the bite.

(152) Answer : (2)
Hint:
Spherical bacteria
Solution:
Bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae* cause pneumonia in humans. *Plasmodium vivax* is a protozoan and causes malaria.
Entamoeba histolytica causes amoebiasis.
Rhino virus is the causative agent of common cold.

(153) Answer : (1)
Solution:

Boyer observed that these enzymes had capability of cutting DNA in particular fashion.

(154) Answer : (1)

Hint:

Microneedle is used.

Solution:

The alien gene is introduced into a cell by injecting it in the nucleus of the target cell under a microscope. Biolistics/gene gun method is suitable for plants. Short impulses of high field strength are given in electroporation.

(155) Answer : (3)

Solution:

In human beings, after one month of pregnancy, the embryo's heart is formed. By the end of the second month of pregnancy, the foetus develops limbs and digits. By the end of 12 weeks (first trimester), most of the major organ systems are formed, for example, the limbs and external genital organs are well developed. By the end of about 24 weeks (end of second trimester), the body is covered with fine hair, eye-lids separate, and eyelashes are formed.

(156) Answer : (2)

Solution:

Chlorophyll *a* is the primary photosynthetic pigment. Chlorophyll *b* accounts for about one fourth of total chlorophyll content and appears yellow-green in chromatogram. Xanthophyll appears yellow in the chromatogram. Carotenoids protect the plant from excessive heat and prevent photo-oxidation of reaction centre.

(157) Answer : (1)

Hint:

Interzonal fibres are seen between two separating chromatids.

Solution:

During anaphase of mitosis, the formation of interzonal fibres occur.

(158) Answer : (4)

Solution:

The trait considered in the pedigree chart is due to autosomal recessive gene.

The trait in an individual regarding recessive gene will be shown only in homozygous condition. In the given pedigree chart, the couple F_1(I) and (II) are homozygous recessive. Therefore, all their children will be affected.

(159) Answer : (4)

Solution:

The plumule and radicle are enclosed in sheaths which are called coleoptile and coleorhiza respectively.

(160) Answer : (3)

Solution:

Microfilaments help in pseudopodia formation. Microtubules form the cytoskeleton of cilia and flagella. Centrioles help in the formation of basal bodies. Intermediate filaments are involved in formation of scaffolds.

(161) Answer : (4)

Solution:

ABO blood groups are controlled by the gene I, has three alleles I^A , I^B and i.

(162) Answer : (1)

Hint:

The dissolution of the synaptonemal complex takes place in the diplotene stage of meiosis I.

Solution:

The correct sequential order for the stages and events is as follows:

d → b → a → e → c

(163) Answer : (4)

Solution:

In angiosperms, egg apparatus consists of two synergids and one egg cell.

(164) Answer : (3)

Hint:

Many *Citrus* varieties show polyembryony.

Solution:

In many *Citrus* varieties some of the nucellar cells surrounding the embryo sac start dividing, protrude into the embryo sac and develop into the embryos. In such species each ovule contains many embryos.

(165) Answer : (3)

Solution:

1. Streptokinase is produced by the bacterium *Streptococcus*. Which is modified by genetic engineering and used as a clot buster.

(166) Answer : (1)

Solution:

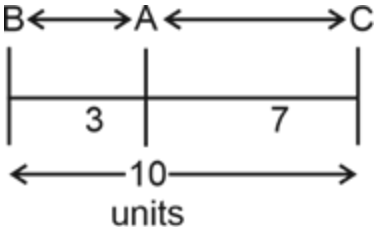
Lac operon are has three structural genes, namely *lac z* , *lac y* and *lac a* .

(167) Answer : (4)

Hint:

More map unit means greater the distance between genes.

Solution:



(168) Answer : (1)

Solution:

Griffith concluded that R strain bacteria had somehow been transformed by heat-killed S-strain bacteria, which had enabled R strain to synthesise a smooth polysaccharide coat and become virulent.

(169) Answer : (2)

Solution:

Sickle cell anaemia	– Defect is caused by transversion of gene controlling β-chain of haemoglobin
α-Thalassemia	– Controlled by two closely linked genes HBA1 and HBA2 on chromosome 16
Phenylketonuria	– Inborn error of metabolism
Haemophilia	– Due to presence of defective form of blood clotting factor

(170) Answer : (3)

Solution:

If the sequence of mRNA is 5' CACUGCAUGUGGGAG 3' then the sequence of the template strand will be 3' GTGACGTACACCCTC 5'

(171) Answer : (3)

Solution:

Conventional methods like urine and serum analysis are not helpful for early detection of a disease.

(172) Answer : (3)

Solution:

*Hind*II recognises a recognition sequence of 6 base pairs and is isolated from the bacteria of genus '*Haemophilus*.

It cuts the DNA helix at specific site in sugar-phosphate backbone'.

(173) Answer : (3)

Solution:

Insulin produced by biotechnology is effective, non allergenic, has same structure that natural one.

(174) Answer : (4)

Solution:

Duramater is the outer most cranial meninx. It is thick and tough layer made up of a tough connective tissue called fibrous tissue.

(175) Answer : (3)

Solution:

In ICSI a single sperm is directly injected into the ovum.

(176) Answer : (3)

Solution:

The cutting of DNA by restriction endonuclease results in the fragments of DNA. These fragments can be separated by a technique known as gel electrophoresis.

(177) Answer : (1)

Solution:

GM plants are more resistant to abiotic stresses

(178) Answer : (4)

Solution:

Human male ejaculates about 200 to 300 million sperms during a coitus.

(179) Answer : (4)

Hint:

Most of the mammals have enucleated RBCs

Solution:

RBCs are nucleated in frogs and contain red coloured pigment namely haemoglobin. The hind-brain of frogs consists of cerebellum and medulla oblongata. Urethra is the common urinogenital duct in human males. Renal portal system is absent in humans.

(180) Answer : (3)

Hint:

Neurons are polarised during the resting state.

Solution:

When a neuron is not conducting any impulse *i.e.,* resting, the axonal membrane is comparatively more permeable to K^+ and nearly impermeable to Na^+. Similarly, the membrane is impermeable to negatively charged proteins present in the axoplasm.

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